

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Induction Bend
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006
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Specification for Induction Bend

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Revision History

Rev No.	Date	Description of Revision
R01	17.05.22	First Issue for Review
A01	13.12.22	Issued for Approval

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipeline Wall Thickness Calculations
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006	Data Sheets for Induction Bend
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Specification for Pipeline Cleaning ,Gauging and Hydrotesting
DEP 31.40.00.10-Gen	Pipeline Engineering (Amendments/Supplements To ISO 13623)
DEP. 82.00.10.10-Gen	Project Quality Assurance

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

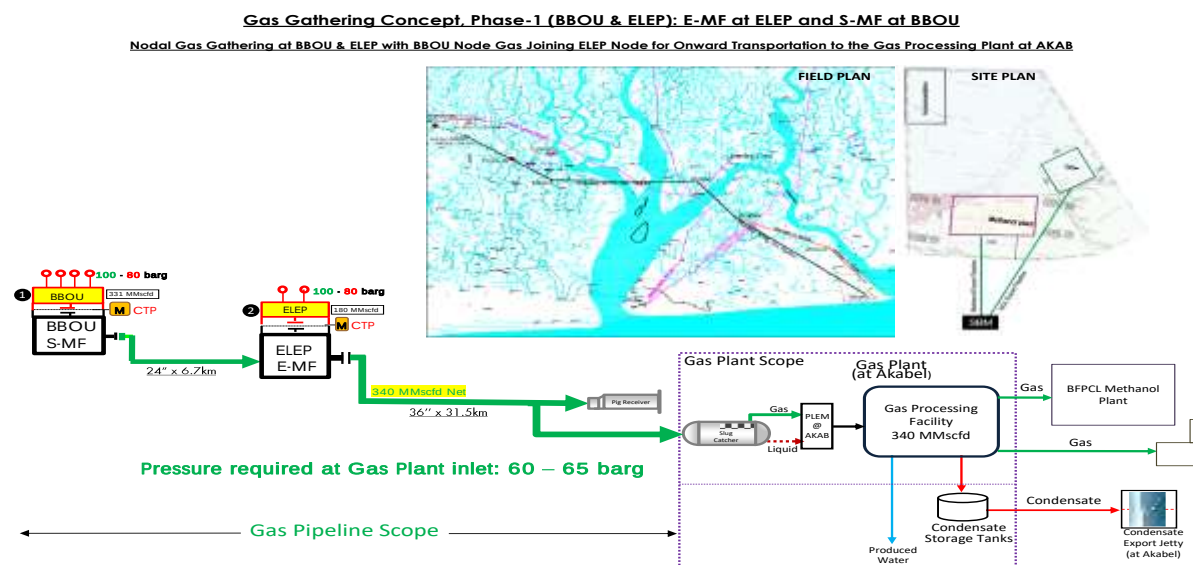


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this specification is to provide the minimum requirements that relate to the design, manufacture, and supply of carbon steel pipeline induction bends for used to achieve connectivity for phase 1 of BFPCL Gas Gathering Pipelines & Associated Infrastructure.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
API 5L	Specification for Line Pipe
API 1104	Welding of Pipelines and Related Facilities
API RP 5L1	Recommended Practice for Railroad Transportation of Line Pipe
API RP 5LW	Recommended Practice for Transportation of Line Pipe on Barges and Marine Vessels
ASME B16.49	Factory-Made, Wrought Steel, Buttwelding Induction Bends for Transportation and Distribution Systems
ASME B31.8	Gas Transmission and Distribution Piping Systems
BS 8010-2.8	Code of Practice for Pipelines. Pipelines on land: design, construction and installation.
BS EN 10204	Metallic Products – Types of Inspection Documents
ISO 9001	Quality Systems— Model for Quality Assurance in Design, Development, Production, Installation and Servicing.
ISO 3183	Petroleum and Natural Gas Industries — Steel Pipe for Pipeline Transportation Systems
ISO 3183-3	Petroleum and Natural Gas Industries -- Steel Pipe for Pipelines -- Technical Delivery Conditions – Part 3: Pipes of Requirement Class C
ISO 15590-1	Petroleum and Natural Gas Industries – Induction Bends, Fittings and Flanges for Pipeline Transportation Systems – Part 1: Induction Bends
ISO 13665	Seamless and Welded Steel Tubes for Pressure Purposes - Magnetic Particle Inspection of the Tube Body for the Detection of Surface Imperfections
ISO 13623	Petroleum and Natural Gas Industries -- Pipeline Transportation Systems

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1 Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
HAZ	Heat affected zone
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
MPS.	Manufacturing Procedure Specification
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	24"
Ultimate Tensile Strength	570 MPa
Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate /hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipe-	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield	483 Mpa
NPS	36"
Ultimate Tensile Strength	570MPa
Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coat-	3LPE
Cathodic Protection	Impressed current

4. General Requirements

4.1. General

This Specification covers the manufacture of bends using the automatic induction heating and water quenching techniques.

4.2. Pipe

The contractor shall be responsible for supplying pipe material suitable for manufacturing bends in accordance with this Specification and Line Pipe Specification

4.3. Bend Wall Thickness

The straight mother pipe shall be selected such that after bending the wall thickness at the extrados shall be no less than the specified minimum. After bending the wall thickness at the extrados shall be within the tolerances specified for the straight mother pipe.

4.4. Tangent Length

All bends shall be supplied with tangent lengths of 500 mm, unless otherwise specified by EMPLOYER.

4.5. Bend Radius

All bends shall be supplied with **5D** bend radius, unless otherwise specified by EMPLOYER.

4.6. Control of Induction Bend Heat Input

Induction bending shall be carried out in the temperature range of 850°C to 1060°C. The whole pipe length, both bend and tangent ends, shall be subject to the same induction heating cycle and post bending heat treatment unless agreed otherwise by EMPLOYER.

The temperature shall be controlled within $\pm 15^{\circ}\text{C}$ with a permanent installed measuring device such as a pyrometer which is calibrated against a thermocouple temperature measuring instrument.

4.7. Post Bend Heat Treatment

The heat treatment process shall be performed in accordance with a documented procedure which shall be supplied at the same time as the Manufacturing Procedure Specification (MPS). Post-bend heat treatment shall be over the entire bend and tangent lengths. Full details of heat treatment procedures, including furnace layout, thermocouple locations, heating times, quenching temperatures and soak times, cooling medium and rate shall be supplied to the EMPLOYER for review and approval. No bends shall be stacked in furnace. Bends shall be spaced evenly on furnace base to allow complete air circulation and even heating/cooling.

All post-forming heat treatments shall use attached thermocouples with calibrated chart recorders.

5. Quality Assurance And Quality Control

The supplier shall show that an effective quality assurance system is in operation for both product and services and this shall comply with ISO 9001 or equivalent as approved by EMPLOYER. The supplier shall establish and maintain a fully documented Quality Assurance / Quality Control system in accordance with ISO 9001 to ensure:

- Effective inspection and objective evidence that items conform to Purchase Order requirements
- Adequate identification and suitable handling of items.

The supplier shall submit on request, documentary evidence of any Quality Reviews to ISO 9001 and internal system audits and corrective action requests.

All certification, test results, reports and any other documentation submitted to the EMPLOYER relative to the inspection system and results of inspections shall be in the English language.

Any verification of quality performed by EMPLOYER or its representative will not relieve the supplier in any way of its responsibility to produce acceptable items nor shall it preclude any subsequent rejection of items.

EMPLOYER shall establish mandatory hold points where applicable, which require verification of selected characteristics of any items or process by EMPLOYER and beyond which the manufacture shall not proceed until released. The supplier shall establish and maintain a system of traceability, which complies with ISO 9001. The system shall be such that each line-pipe is allocated a unique number, and this number shall be maintained such that at any point the individual line-pipe can be traced back to its original heat number. These numbers will be in blocks allocated by the EMPLOYER to the supplier and the supplier shall demonstrate its ability to maintain traceability of such numbers throughout production.

6. Designation

Designation of fittings shall comply with ISO 15590-1:2009 (Section 6) and shall be represented as "IB-483-PSL 2" for the induction bends.

Where:

- PSL 2" identify the technical delivery conditions class for induction bends in non-sour service
- 483 is the specified minimum yield strength, expressed in mega Pascal (MPa)

7. Pressure Rating And Design

The hoop stress in the induction bend due to internal fluid pressure shall not exceed the hoop stress permitted in ISO 13623, or other applicable design code, for straight pipe in the location of the bend. Compliance with this requirement shall be demonstrated either by calculations or by satisfying both of the following requirements.

- The wall thickness of the bend extrados shall be at least t_{min} .
- The wall thickness at the bend intrados (inside radius) shall be at least as given in Equation below:

$$t_i = t_{min} \frac{2r_b - r_p}{2(r_b - r_p)}$$

Where:

r_p = nominal outside diameter

r_b = bend centre line radius

t_{min} = nominal design wall thickness

t_i = minimum required thickness at the intrados.

Note: The detailed calculation report of minimum required thicknesses at the intrados and extrados including the drawings shall be issued to EMPLOYER for approval before commencement of manufacturing.

For pipelines not designed in accordance with ISO 13623, the wall thickness of the bend extrados (outer radius) may be less than min.

The requirements in this clause address the design of a bend against internal pressure. It is necessary that the designer also consider other loads, both static and dynamic, and pipeline test conditions to demonstrate compliance with the strength requirements of ISO 13623 or any applicable other applicable design code. ISO 15590-1 should be consulted for additional load requirements to be considered in the design of induction bends. It is also critical that ovality of the straight sections adjacent the bends should not exceed 2.5 % to avoid in-service failure due to collapse (from soil cover weight).

8. Manufacturing

8.1. Mother Pipe

Mother pipe shall be manufactured as within the requirements as stated in the DEP 31.40.20.37 and line pipe specifications report. The carbon steel shall be ISO 3183, PSL 2.

EMPLOYER shall be notified if there is intent to upgrade or modify the strength of either the mother pipe or induction bend after the bending process.

8.2. Qualification Test Bend

The test bends shall be manufactured with approved MPS as stated in the DEP 31.40.20.33 Gen

8.3. Production Bending

Induction bending shall be carried out in accordance with the qualified MPS. Interruption of the induction bending operation shall result in rejection of the bend.

8.4. Post-Bending Heat Treatment

Except acceptable properties can be demonstrated after bending for any L483 X70 line, the Contractor shall provide a quench and temper heat treatment of the entire joint.

The temperature of each furnace-load of bends shall be monitored by thermocouples connected directly to selected bends and shall be recorded. The type and location of the thermocouples shall be as specified in the MPS.

8.5. Strip/Plate End Welds

Induction bends shall not contain coil-strip end welds or plate end welds.

8.6. Forming and Sizing after Bending

This shall be done according to ISO 15590-1 as amended by DEP 31.40.20.33- Gen.

8.7. Jointers and Girth Welds

Induction bends ordinarily should not contain girth welds. The DEP 31.40.20.33-Gen permits jointers during manufacture of bends on the condition that on completion of the bend, the joiner shall be removed such that no effect from the joiner shall remain including the heat affected zone.

8.8. End Preparation

Bends shall be square except otherwise specified by EMPLOYER. The cutting method shall not affect the properties of the bend when compared to the mother pipe.

9. Testing And Inspection

9.1. General

Inspection and Testing shall be carried out in accordance with Section 9 of ISO 15590-1 or as amended by DEP 31.40.20.33-Gen. All procedures proposed for non-destructive testing shall be submitted for EMPLOYER approval.

Prior to the commencement of manufacture, the supplier shall prepare a written Inspection and Test Plan, which describes the inspection to be performed on each product. The Inspection and Test Plan, reference procedures and changes thereto are subject to the approval of the EMPLOYER.

Supplier shall provide the following in addition to the Inspection and Test Plan:

- A flowchart illustrating each manufacturing point affecting the quality of the product and its relative location in the production cycle where conformance of characteristics shall be verified. The supplier shall include all inspection points of its own verification of quality.
- The characteristics to be inspected at each inspection point from procedural approval to shipping inspection. The procedures shall be submitted to EMPLOYER for review and approval prior to production commencement.

9.2. Quality Records

The supplier shall maintain quality records of all tests, inspection and measurements detailed in this document and the EMPLOYER approved quality plan as documentary evidence of conformance to quality requirements. Quality records shall be made available to EMPLOYER's representative for analysis and review.

Quality records shall include item identification by reference to revision number, acceptance criteria, specific inspection performed and results obtained (If measurements are not a requirement, include in the record the basis of acceptance), date of inspection, identification of inspector, data recorder charts, qualification of material, personnel, procedures and equipment.

The supplier shall perform trend analyses of all factors affecting the quality of the product.

9.3. Final Inspection

The supplier shall inspect the final item to ensure compliance with contract requirements. A check shall be made of all inspection records to verify that items were inspected at all points shown in the Inspection and Test Plan and that these records are complete and available to EMPLOYER's representative.

Visual inspection and dimensional checks shall be performed both at the point of discharge and receipt to confirm that no damage took place during transportation.

9.4. Verification

All supplier inspection systems may be subject to evaluation and surveillance by EMPLOYER's representative to ensure that they meet the requirements of this document and Purchase Order. The supplier's operations required by this standard are subject to;

- Procedure compliance checking at unscheduled intervals to determine that the supplier's inspection system is being effectively applied.
- Product verification to determine compliance with contract requirements. The method of verification is at the discretion of EMPLOYER's representative.

Inspection Certification shall be in accordance with BS EN 10204 3.2. All certifications and qualifications shall be certified by an independent certifying authority, such as Lloyds, DNV, etc.

10. Accommodation and Assistance

The supplier shall provide EMPLOYER's representative with office accommodation and facilities required for the proper accomplishment of his work and shall provide any reasonable assistance required by EMPLOYER's representative for verification, documentation, or movement of items.

EMPLOYER's representative shall be afforded unrestricted opportunity to perform verification of inspection. The supplier shall make available its inspection equipment for use by the EMPLOYER's representative to verify conformance to contract requirements.

The supplier personnel shall be made available for the operation of inspection equipment as reasonably required by EMPLOYER.

11. Defects

11.1. Repair of Defects

The EMPLOYER shall be advised of each bend that is designated for rectification / repair work. Supplier shall issue a repair procedure to be approved by EMPLOYER before production works commences.

11.2. Reject Bends

The EMPLOYER shall be advised of all bends that have been rejected and taken away from the original stockpiles.

Rejected bends shall be stored on special designated marked areas and must be subject to NCR / CAR documents and the EMPLOYER shall be notified in writing.

12. Surface Finish

All bend surfaces, both internally and externally, shall be grit blasted and shall be free from oil, grease and other contaminants that might interfere with coating application. The acceptance criteria for surface roughness of bends shall be agreed between the Contractor and the EMPLOYER prior to the start of production.

No lacquer shall be applied.

13. Documentation

All certification and documentation shall be as per ISO 3183 certification requirements except as amended by DEP 31.40.20.37.Gen.

The supplier shall deliver materials with certificates from a classification society in accordance with BS EN 10204 3.2. The certificates and associated documentation supplied to EMPLOYER shall contain all relevant information, including, but not restricted to, the following:

- Supplier's Identification
- EMPLOYER's Name and Order Number
- Metallographic Examination Records
- Description of Product, Dimension, Identification Number, Weight
- Steel Making Process
- Chemical Composition
- Heat Number
- Identification of Steel type and Grade
- Yield - Tensile Ratio
- Charpy Impact Results
- Hardness Tests
- Heat Treatment Condition and Records
- Hydrostatic Test
- NDE Procedures and Results
- Surface Inspection
- Dimensional Control Checks
- Manufacturing Procedure and Qualification Tests
- EMPLOYER's Induction Bend Release Note.
- Supplier's Third Party BS EN 10204 3.2 Certificates

The supplier shall also supply a table of information stating the heat number, length, weight, testing and inspection associated with each bend number.

14. Marking And Protection Of Bends

14.1. Marking

Marking shall be in accordance with ISO 15590-1, section 12. In addition, the words —DEP 31.40.20.33.Gen shall be included in the stencil area. Markings shall be made with indelible paint on the inside surface or, if it is not possible to mark on the inside surface, then on the outside for smaller diameter bends.

Markings shall be executed in block capitals with a minimum height of 19 mm (0.748 in). Identification markings shall not be stencilled or painted on the weld preparation.

Both ends of the bend shall be marked with the following information:

- Manufacturer's name or trade mark;
- Number of this part of ISO 15590-1 Ref [20]
- Pipe material grade
- Purchase order and item number
- Bend designation
- Diameter (outside or inside)
- Minimum nominal wall thickness
- Bend designation
- Bend angle
- Bend radius
- Heat number or manufacturer's heat identification
- Unique bend number;
- Any additional marking specified in the purchase order

14.2. Coating

The external of bend shall be protected using a 3 layer Polyethylene coating (3LPE) as per Technical Specification for Line Pipe Coating.

14.3. Bevel Protectors/End Caps

Unless otherwise specified by the EMPLOYER, bends shall be supplied with bevel protectors or end caps firmly secured by tape. The method of bend end protection shall be subject to approval by the EMPLOYER.

15. Transportation

15.1. General

The supplier shall submit to the EMPLOYER for review and approval, information detailing the proposed method of handling, stacking during storage, and stacking and securing bends for transportation.

No welding of temporary attachments for handling, stacking, or securing shall be permitted. All handling, loading and unloading shall be performed in such a manner as to avoid mechanical damage and corrosion.

15.2. Bevel Protectors

Bevel protectors of a type approved by the EMPLOYER shall be supplied and installed by the supplier before stockpiling of the bends takes place.

15.3. Loading and Stacking

Rail cars, trucks, lighters, ships or other conveyances shall be cleaned of debris and any substance that might damage the induction bends prior to loading. Suitable timber cradles or other packing shall be used to protect the induction bends and coating if applicable against damage in transit. Loading on or into rail cars, trucks, lighters, ships, or other conveyance shall be performed in accordance with API RP 5L1 and API RP 5LW as appropriate. Stacks shall not exceed 4 meters in height. In all cases loading shall be in accordance with a procedure approved by the EMPLOYER.

When the induction bends are transported by truck, the number of induction bends transported per truck at any one time shall be limited to 6, unless the supplier can demonstrate that it is safe to transport more safely and duly approved in writing by EMPLOYER. In addition to securing the induction bends on the truck to prevent sideways movement, provision shall be made to prevent longitudinal movement to resist forces due to the vehicle braking, accelerating or in negotiating an incline.

Loading and unloading of induction bends shall be witnessed by EMPLOYER's representative whose decision on the suitability of protection / handling is final.

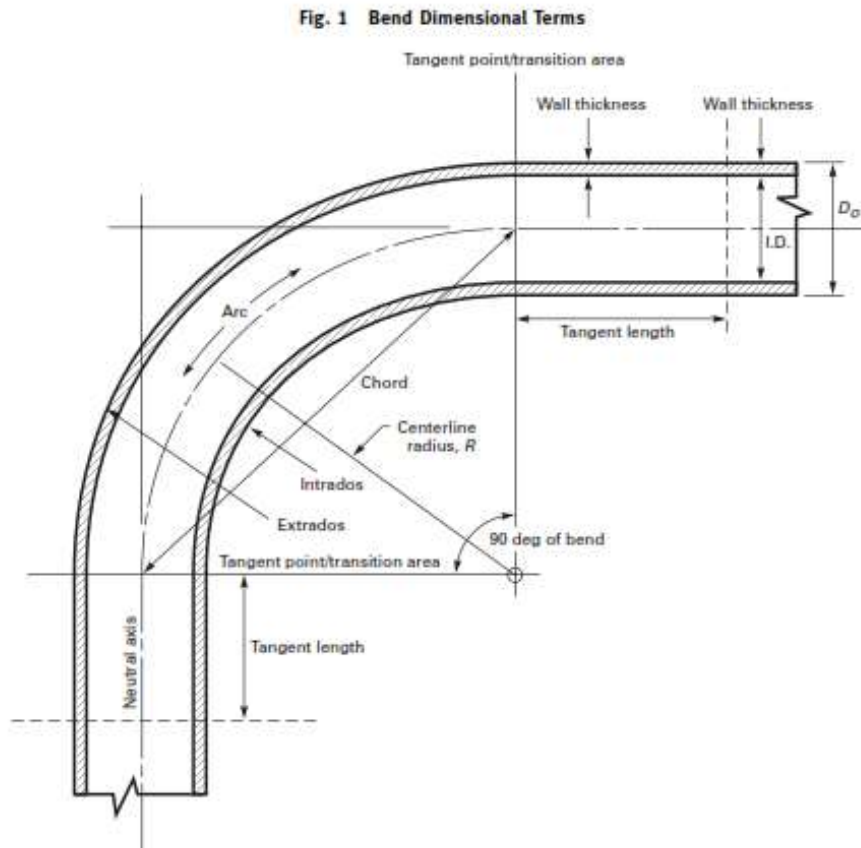
15.4. Protection

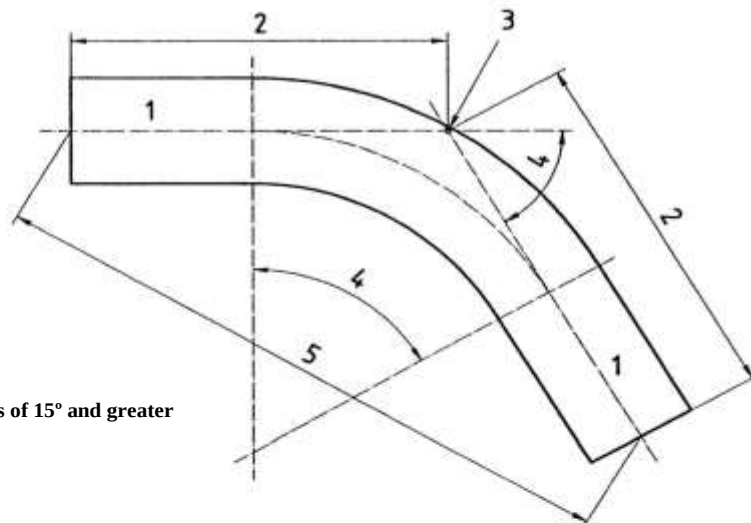
The supplier shall propose to the EMPLOYER for approval, suitable methods for the protection of internal surfaces against moisture during transportation. Finished bends to be stored for a significant period at the bending yard shall be stored in a manner that prevents corrosion. Induction bends shall not rest on projections, which could result in point stresses or be allowed to rub on an adjacent object. Loading and stacking shall be limited to prevent stresses, which result in out-of-roundness.

No Induction bend shall be shipped above deck on sea going open barges or in such a manner that it is exposed to salt water spray or atmosphere. If it is determined that the delivered bend is chloride contaminated, the bend supplier will reimburse EMPLOYER for the cost of removing the chloride contamination.

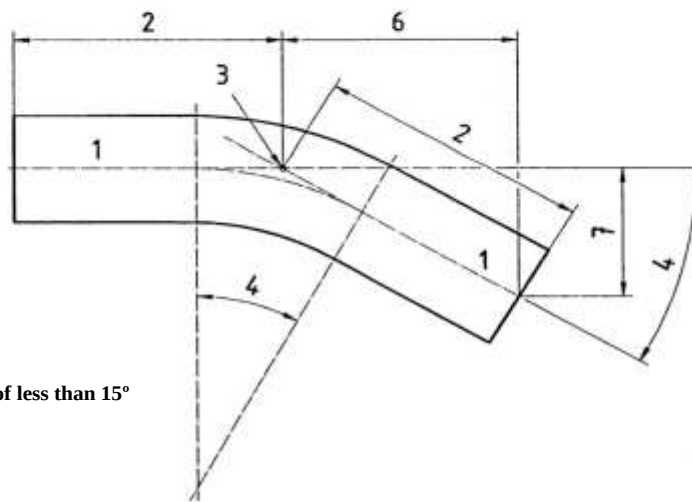
16. Attachments

16.1. Appendix 1





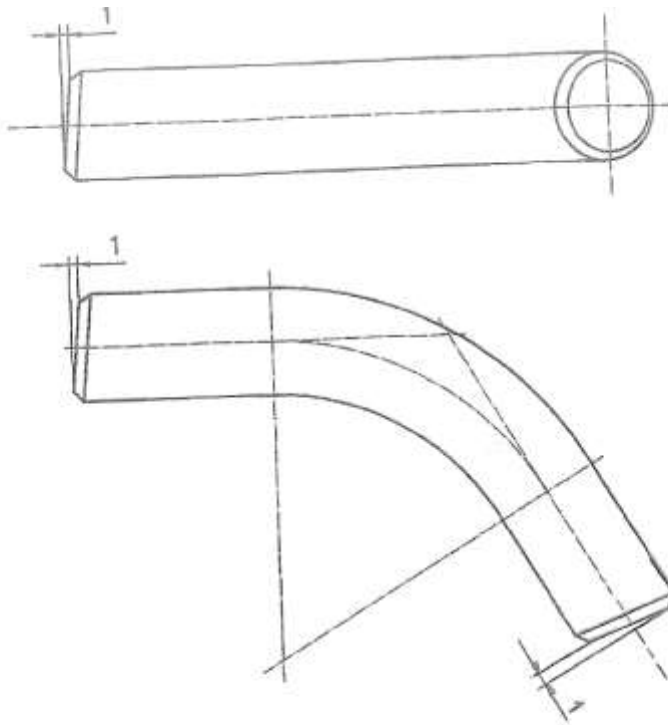
a) Bends with angles of 15° and greater



b) Bends with angles of less than 15°

- a. Key:
- b. 1. Centreline axis
- c. 2. Centre to end
- d. 3. Centre of bend
- e. 4. Bend angle
- f. 5. Chord
- g. 6. Extension
- h. 7. Offset

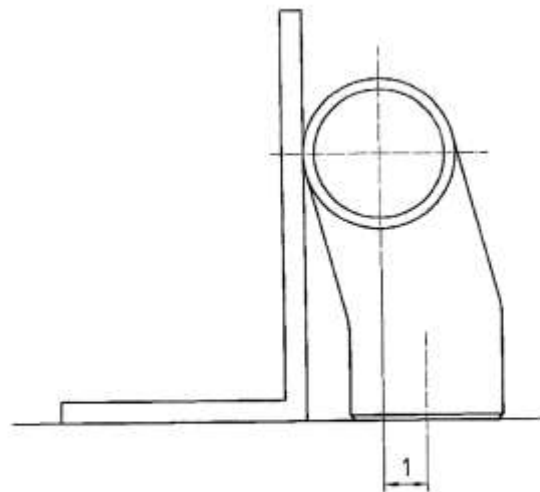
Determination of End Out-of-Squareness



Key

1 Out-of-squareness

Determination of Out-of-Plane



Key

1 Out-of-planeness